Dispersion

September 21, 2022

As you come in...

1. Find the "hericanedata.csv" file on CANVAS
2. Download it, and save it to your PLSC 309 Folder (if you haven't already)
   - If you have a Mac, save as a .csv file, not a "numbers" file.

Measures of Central Tendency

- Mean
- Median
- Mode

Find the Mean

$$X = [5,6,7,8,9]$$

$$Z = [1,6,7,8,13]$$

Find the Mean

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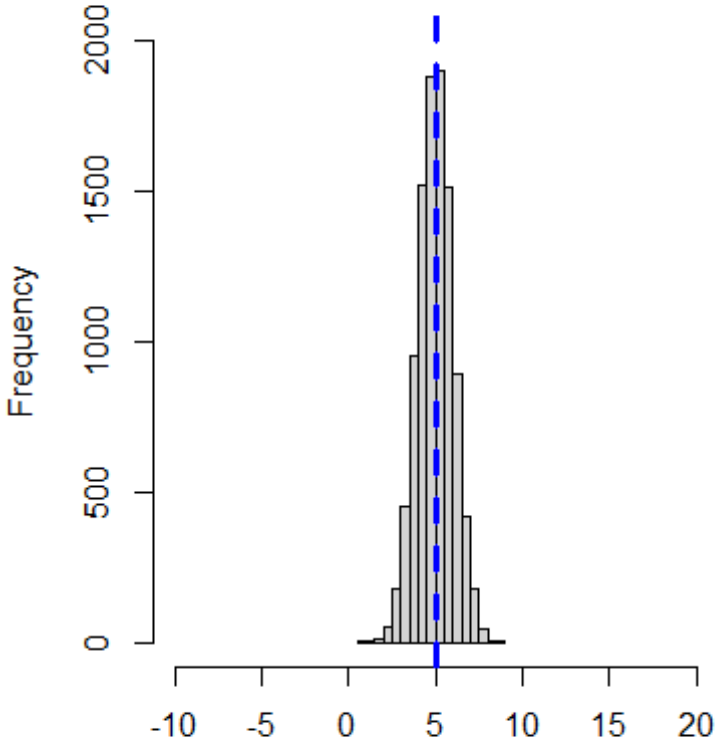
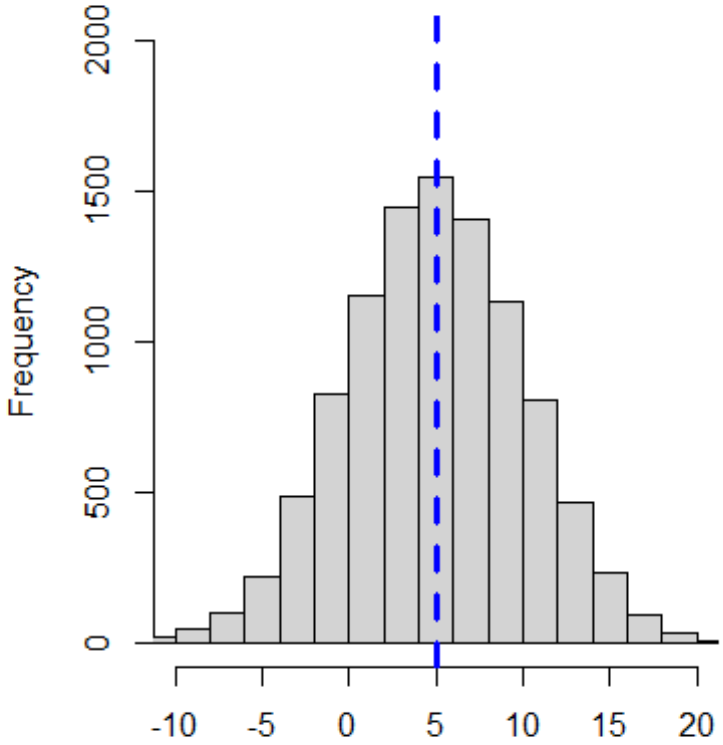
$$Z = [1,6,7,8,13]$$

What's the difference between X and Z?

Dispersion (Conceptually)

How "spread out" are the data relative to the most central value?





Frequency Distribution (Again)

How spread out are these data?

Score	Frequency	Percent
10	1	1/15=6.67%
16	1	1/15=6.67%
17	1	1/15=6.67%
18	1	1/15=6.67%
20	1	1/15=6.67%
24	2	2/15=13.33%
27	1	1/15=6.67%
28	2	2/15=13.33%
31	1	1/15=6.67%
35	1	1/15=6.67%
38	1	1/15=6.67%
44	1	1/15=6.67%
46	1	1/15=6.67%
	15 games	**100%**

Percentiles

Percentile: the percentage of observations with values less than or equal to that number

- Median is the observation at which 50% of the rest of the data fall below it

Score	Frequency	Percent	Cumulative Perc.
10	1	1/15=6.67%	6.67%
16	1	1/15=6.67%	13.33%
17	1	1/15=6.67%	20%
18	1	1/15=6.67%	26.67%
20	1	1/15=6.67%	33.33%
24	2	2/15=13.33%	46.67%
27	1	1/15=6.67%	53.33%
28	2	2/15=13.33%	66.67%
31	1	1/15=6.67%	73.33%
35	1	1/15=6.67%	80%
38	1	1/15=6.67%	86.67%
44	1	1/15=6.67%	93.33%
46	1	1/15=6.67%	100%
	15 games	**100%**	

Range

Distance between the minimum and maximum value

- Minimum: smallest value
- Maximum: largest value

Interquartile Range

The middle 50% of the data:

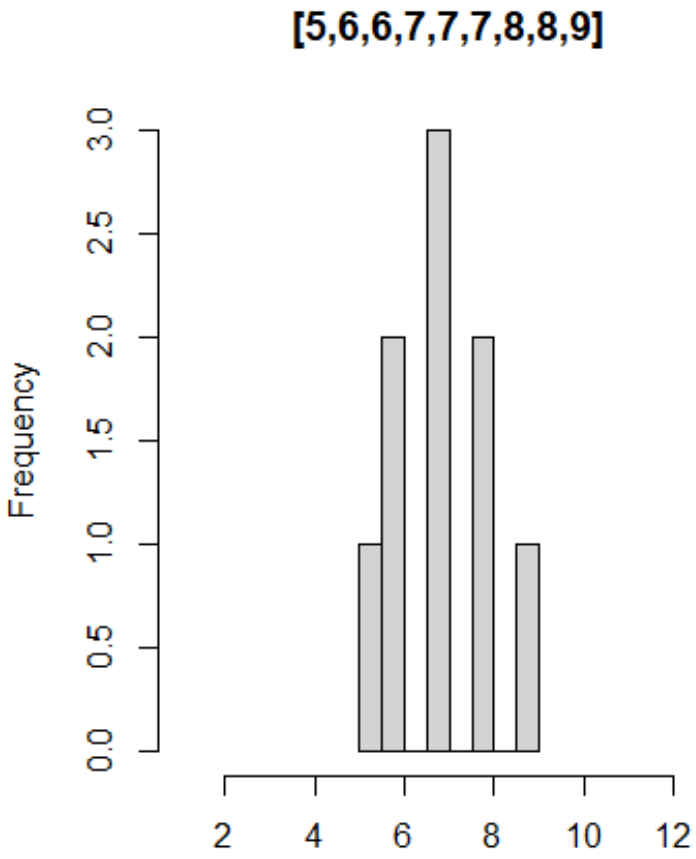
- 75th Percentile - 25th Percentile

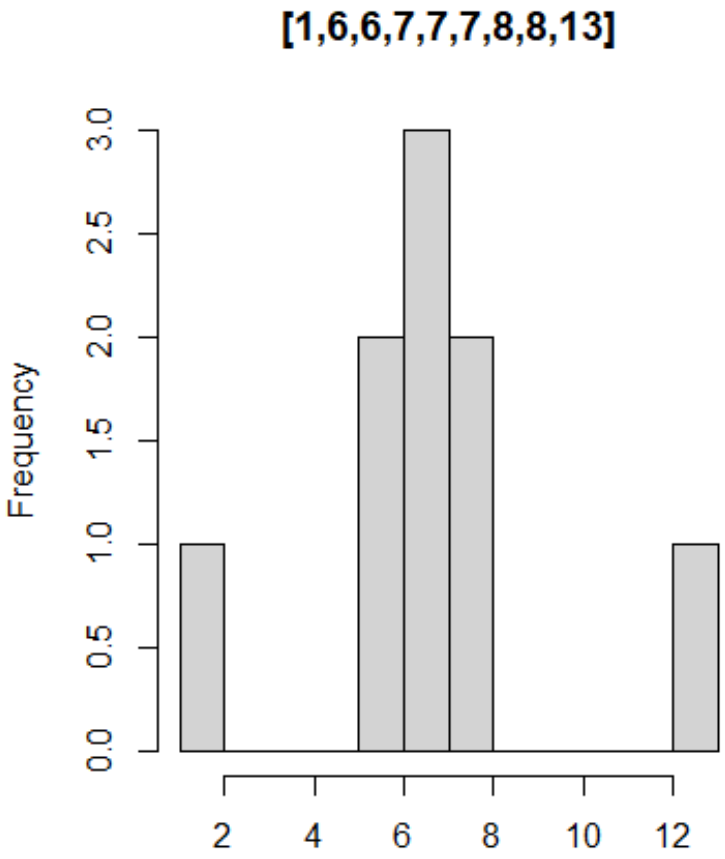
Upsides:

- Simple, intuitive
- Rank statistic, like the median, so resistant to outliers!
- Helps us <u>formally</u> identify outliers
  - Outlier if value $> 1.5x(IQR)$ <u>more</u> than 75th percentile or $< 1.5x(IQR)$ <u>less</u> than 25th percentile

Identical IQR of Different Distributions

Downside of IQR: Does not use all data in distribution





The Logic of Variance

Let's say we want to summarize the spread of a distribution with one number and use *all* of the data:

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   - Example: mean of [1,2,3,4,5] is 3. First observation deviates from the mean by $3 - 1 = 2$.

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   - Example: mean of [1,2,3,4,5] is 3. First observation deviates from the mean by $3 - 1 = 2$.
3. We can calculate those deviations for every observation in variable X,
4. and then take (approximately) the "average" of those deviations!
   - Sum over all deviations, then divide by sample size...

The Logic of Variance

Two problems though...

1. Some deviations will be positive (observations greater than mean), some will be negative (observations smaller than mean).
   - In a symmetrical distribution, the average distances will be 0!
   - Solution: Square the deviations before summing them. Then, summation will always be > 1

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2. Sample deviations tend to be slightly closer to mean than the average deviation in a *population*.
   - Solution: Divide by $n - 1$ rather than n

Variance

The sum of squared differences between each value and its mean divided by the number of cases minus 1.

$$s^2 = \frac{\sum(X - \bar{X})^2}{n - 1} \tag{1}$$

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But interpretation is funky... we want to know the average deviations, not the average squared deviations.

Calculating Variance

$$s^2 = \frac{\sum(X - \bar{X})^2}{n - 1}$$

1. Find the mean of X
2. Take difference of each observation from $\bar{X}$
3. Square each deviation
4. Add up all these squared deviations
5. Divide by $n - 1$

Calculating Variance of PSU Scores

Number	Opponent	PSU Score	Mean Score ($\bar{x}$)	$\bar{x} - x_i$	$(\bar{x} - x_i)^2$
1	Wisconsin	16	27.07		
2	Ball State	44	27.07		
3	Auburn	28	27.07		
4	Villanova	38	27.07		
5	Indiana	24	27.07		
6	Iowa	20	27.07		
7	Illinois	18	27.07		
8	Ohio State	24	27.07		
9	Maryland	31	27.07		
10	Michigan	17	27.07		
11	Rutgers	28	27.07		
12	MSU	27	27.07		
13	Arkansas	10	27.07		
14	Purdue	35	27.07		
15	Ohio	46	27.07		
	Total	406			

Calculating Variance of PSU Scores

Number	Opponent	PSU Score	Mean Score ($\bar{x}$)	$\bar{x} - x_i$	$(\bar{x} - x_i)^2$
1	Wisconsin	16	27.07	11.07	122.5449
2	Ball State	44	27.07	-16.93	286.6249
3	Auburn	28	27.07	-0.93	0.8649
4	Villanova	38	27.07	-10.93	119.4649
5	Indiana	24	27.07	3.07	9.4229
6	Iowa	20	27.07	7.07	49.9849
7	Illinois	18	27.07	9.07	82.2649
8	Ohio State	24	27.07	3.07	9.4249
9	Maryland	31	27.07	-3.93	15.4449
10	Michigan	17	27.07	10.07	101.4049
11	Rutgers	28	27.07	-0.93	0.8649
12	MSU	27	27.07	0.07	0.0049
13	Arkansas	10	27.07	17.07	291.3849
14	Purdue	35	27.07	-7.93	62.8849
15	Ohio	46	27.07	-18.93	358.3449
	Total	406			**1510.9335**

Calculating Variance of PSU Scores

$$s^2 = \frac{\sum(X - \bar{X})^2}{n-1} = \frac{1510.9335}{15-1} =$$

Calculating Variance of PSU Scores

$$s^2 = \frac{\sum(X - \bar{X})^2}{n-1} = \frac{1510.9335}{15-1} = 107.92$$

Standard Deviation

Average difference between all the individual values of X and the mean of X.

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Properties:

- Always 0 (no variation) or greater
- Bigger numbers means the dispersion is greater
- Uses all values in a variable which means...

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$$s = \sqrt{\frac{\sum(X - \bar{X})^2}{n - 1}} \tag{2}$$

Properties:

- Always 0 (no variation) or greater
- Bigger numbers means the dispersion is greater
- Uses all values in a variable which means... sensitive to outliers!

Calculating Standard Deviation

$$s = \sqrt{\frac{\sum(X - \bar{X})^2}{n - 1}}$$

1. Find variance (s^2)
2. Take square root of variance ($\sqrt{s^2}$)

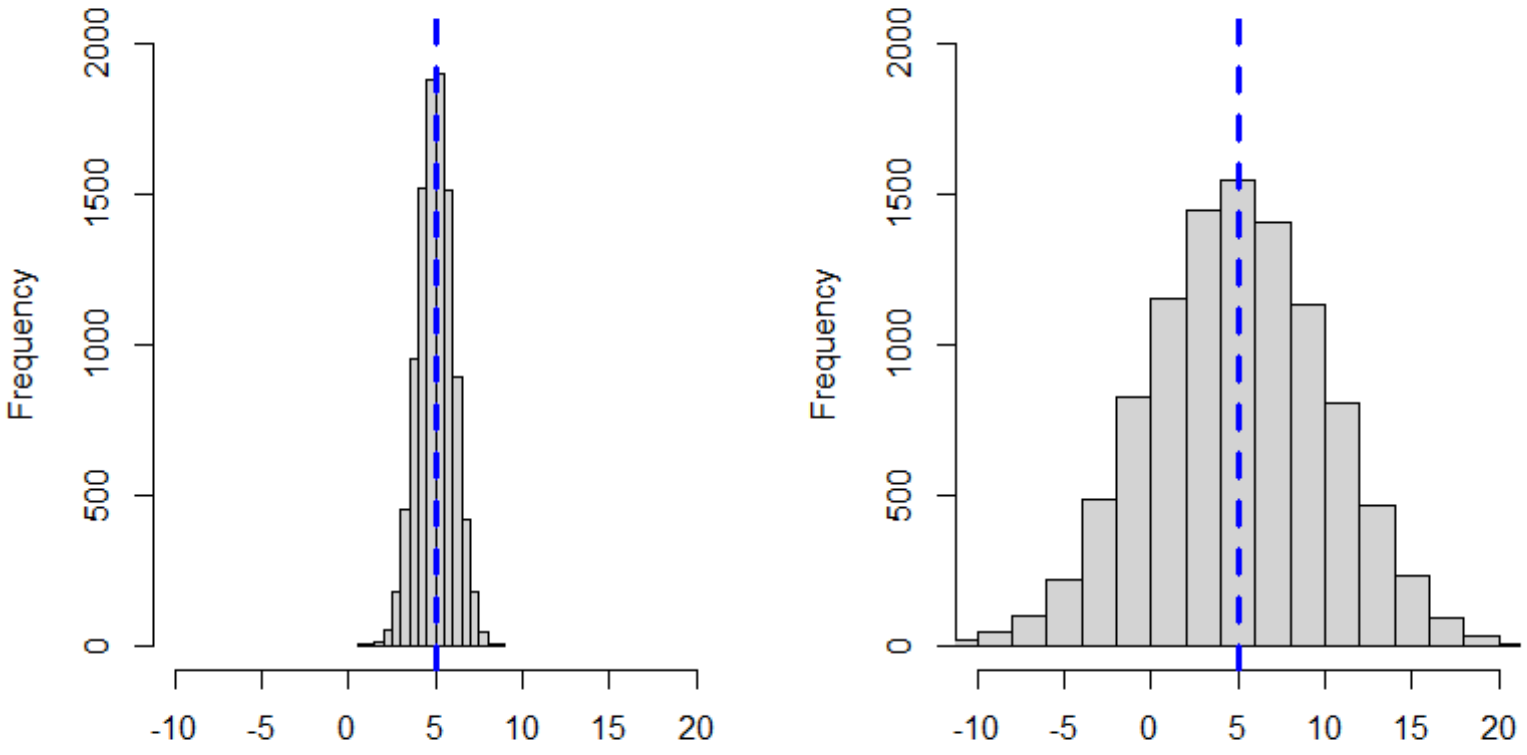
Calculating Standard Deviation of PSU Scores

$$s = \sqrt{\frac{\sum(X - \bar{X})^2}{n-1}} = \sqrt{\frac{1510.9335}{15-1}} =$$

Calculating Standard Deviation of PSU Scores

$$s = \sqrt{\frac{\sum(X - \bar{X})^2}{n-1}} = \sqrt{\frac{1510.9335}{15-1}} = 10.39$$

Distributions with Different Deviations (Again)



Left plot is $s = 1$, right plot is $s = 3$